



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		IV / IV - B.Tech. I - Semester							
MECHANICAL ENGINEERING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23ME4101	CAD/CAM	PC	1	1	0	2	30	70	100
B23ME4102	Artificial Intelligence and Machine Learning	PC	1	1	0	2	30	70	100
#PE-IV	Professional Elective -IV	PE	3	0	0	3	30	70	100
#PE-V	Professional Elective -V	PE	3	0	0	3	30	70	100
#OE-III	Open Elective-III	OE	3	0	0	3	30	70	100
#OE-IV	Open Elective-IV	OE	3	0	0	3	30	70	100
B23ME4113	CAD/CAM Lab	PC	0	0	2	1	30	70	100
B23ME4114	Artificial Intelligence and Machine Learning lab	SEC	0	1	2	2	30	70	100
B23MC4101	Constitution of India	MC	2	--	--	--	30	--	30
B23ME4115	Evaluation of Industry Internship /Mini Project.	PR	--	--	--	2	--	50	50
TOTAL			16	3	4	21	270	610	880

	Course Code	Course
#PE-IV	B23ME4103	Advanced Materials
	B23ME4104	Smart manufacturing
	B23ME4105	Micro Electro Mechanical Systems
	B23ME4106	Functional Materials and Applications
	B23ME4107	12 week MOOC Swayam/NPTEL course recommended by the BoS
#PE-V	B23ME4108	Hydrogen and Fuel Cell Technology
	B23ME4109	Nano Technology
	B23ME4110	Conventional and futuristic vehicle technology
	B23ME4111	Autonomous Systems
	B23ME4112	12 week MOOC Swayam/NPTEL course recommended by the BoS
#OE-III	Student has to study one Open Elective offered by AIDS or AIML or CE or CIC or CSBS or CSG or CSE or CSIT or ECE or EEE or IT or S&H from the list enclosed.	
#OE-IV	Student has to study one Open Elective offered by AIDS or AIML or CE or CIC or CSBS or CSG or CSE or CSIT or ECE or EEE or IT or S&H from the list enclosed.	

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ME4101	PC	1	1	--	2	30	70	3 Hrs.
CAD/CAM								
(For ME)								
Course Objectives: The objective of this course is to								
1.	Impart the fundamentals of CAD/CAM, automation, and computer applications in product design.							
2.	Acquaint the student with computer graphics, geometric transformations and modeling techniques.							
3.	Enable understanding of wireframe, surface, and solid modeling in CAD systems.							
4.	Educate the student on CNC machines and part programming, including APT.							
5.	Provide knowledge on group technology, FMS, and computer-integrated manufacturing systems.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Apply CAD/CAM fundamentals and computer graphics concepts.							K3
2.	Apply the geometric transformations and modeling techniques in CAD.							K3
3.	Analyze CNC machine tools and develop part programs.							K3
4.	Illustrate the concepts of group technology and flexible manufacturing systems.							K3
5.	Demonstrate concepts of CIM, process planning, and quality control.							K3
SYLLABUS								
UNIT-I (6Hrs)	Fundamentals of CAD, CAM, and Automation – Design process – Application of computers in design – Benefits of CAD – Product life cycle of CAD/CAM systems – Comparison of CAD with conventional design methods. Computer Graphics: Raster scan graphics – Coordinate systems – Database structure for graphics modeling – Geometric transformations – 2D & 3D transformations							
UNIT-II (6 Hrs)	Geometric Modeling Techniques: Wireframe Modeling: Definition – Advantages and disadvantages – Wireframe entities – Analytic and synthetic entities. Surface Modeling: Definition – Advantages and disadvantages – Surface entities – Analytic and synthetic surfaces. Solid Modeling: Definition – Constructive Solid Geometry (CSG) – Advantages – Modeling entities. Curve Representation: Implicit and explicit forms of curves – Straight line, circle, ellipse – Cubic spline and Bezier curves – Differences between Bezier and cubic spline curves – Introduction to data exchange formats. Generative AI in CAD.							
UNIT-III (6 Hrs)	Introduction to CAM: Creating Database for manufacturing– Numerical Control (NC): NC modes – NC elements – CNC machine tools – Structure and applications – Machining centers and turning centers – Direct Numerical Control (DNC) – Adaptive control. Part Programming: Fundamentals of CNC programming – Manual part programming methods – Computer-aided part programming – APT language – Simple problems in manual and computer-aided part programming.							

UNIT-IV (6 Hrs)	Group Technology: Part families – Parts classification and Coding – OPTIZ and MICLASS classification systems – Production flow analysis – advantages. Flexible Manufacturing Systems (FMS): Introduction – Types of FMS – FMS Components -Workstations- Material handling & storage systems – Computer Control Systems Equipment – FMS layout configurations.
UNIT-V (6 Hrs)	Computer Integrated Manufacturing Systems (CIMS): Types of manufacturing systems – Machine tools and related equipment – Material handling systems – Material requirement planning (MRP) – Enterprise resource planning (ERP) – Computer control systems – Human labor in manufacturing systems – Benefits of CIMS. Computer-Aided Quality Control: Concepts of quality control – Computer-aided inspection techniques -Contact and Non-contact inspection Techniques Process Planning: Computer-aided process planning – Variant process planning – Generative process planning
Textbooks:	
1.	CAD/CAM-Computer Aided Design & Manufacturing, by M.D. Groover & E.W. Zimmer.
2.	Computer Aided Design and Manufacturing, by Dr. Sadhu Singh, Khanna Publishers
Reference Books:	
1.	Computer Integrated Design and Manufacturing, by David D. Bedworth, Mark R. Henderson & Philip M. Wolfe, McGraw-Hill Book Company, Singapore
2.	Computer Aided Manufacturing, by P.N. Rao, N.K. Tewari & T.K. Kundra, Tata McGraw- Hill publishing company Ltd, NewDelhi
3.	CAD/CAM/CIM by Radhakrishna, New age international
e-Resources	
1.	https://nptel.ac.in/courses/112102102
2.	https://onlinecourses.swayam2.ac.in/e-learning/preview/nou22_me08

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ME4102	PC	1	1	--	2	30	70	3 Hrs.
ARTIFICIAL INTELLIGENCE & MACHINE LEARNING								
(For ME)								
Course Objectives: Students are expected								
1.	To understand the basic concepts of Artificial Intelligence and intelligent agents.							
2.	To learn machine learning fundamentals and supervised learning algorithms.							
3.	To study unsupervised learning techniques and Bayesian methods.							
4.	To understand neural networks and genetic algorithms.							
5.	To gain knowledge of deep learning techniques and machine learning model evaluation.							
Course Outcomes: By the end of the course, the student will be able								
S.No	Outcome							Knowledge Level
1.	Apply the fundamental concepts of Artificial Intelligence and intelligent agents to solve basic problem-solving tasks.							K3
2.	Apply supervised learning algorithms such as regression, decision trees, and support vector machines.							K3
3.	Apply unsupervised learning techniques including clustering and dimensionality reduction.							K3
4.	Apply the principles of Artificial Neural Networks, Deep Learning, and Genetic Algorithms to solve engineering problems.							K3
5.	Apply model evaluation techniques and ensemble learning methods for predictive analysis and optimization in mechanical engineering applications.							K3
SYLLABUS								
UNIT-I (8Hrs)	Introduction to Artificial Intelligence: Definition of Artificial Intelligence, Evolution of AI, Need and applications of AI in real-world engineering, Intelligent Agents, Agents and Environment, Rational Agents, Types of environments, Structure of intelligent agents, Industrial Use cases: Smart Manufacturing (Industry 4.0 basics), Digital Twins, Introduction to Robotics & Automation.							
UNIT-II (8 Hrs)	Introduction to Machine Learning : Definition of Machine Learning, Types of learning, Applications in industry, Regression vs Classification, Performance metrics, Bias and variance, Overfitting and underfitting, Supervised Learning: Linear Regression, Logistic Regression, K-Nearest Neighbour, Decision Trees, Support Vector Machines, Kernel methods.							
UNIT-III (8 Hrs)	Unsupervised Learning: Clustering-Introduction to clustering, K-Means clustering, Applications of clustering, Dimensionality Reduction - Principal Component Analysis (PCA), Kernel PCA. Bayesian Learning: Basic concept and its application, introduction to Naïve Bayes classifier.							

	Computer Vision: Introduction to Computer Vision, basics of image formation and image processing techniques (filtering, edge detection), and applications in manufacturing such as surface defect detection, quality inspection, and visual monitoring systems.
UNIT-IV (8 Hrs)	Neural Networks: Introduction to Artificial Neural Networks, Perceptron and Multilayer Neural Networks (basic concepts). Deep Learning: Introduction to Deep Learning and Convolutional Neural Networks (CNN) with applications in image processing and defect detection. Genetic Algorithms: Introduction to evolutionary computing, basic concepts of selection, crossover and mutation with applications in design optimization.
UNIT-V (8 Hrs)	Model Evaluation and Machine Learning Analytics: Basic model evaluation techniques (accuracy, precision, recall), overfitting and underfitting, model selection concepts. Introduction to ensemble learning methods including Bagging, Boosting, and Random Forest. Applications in Mechanical Engineering: Data-driven applications in experimental fluid mechanics and thermodynamics, predictive modelling for system behavior, predictive maintenance of mechanical systems, and applications in synthesis and optimization of mechanisms.
Textbooks:	
1.	Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, 2/e, Pearson Education, 2010.
2.	Tom M. Mitchell, Machine Learning, McGraw Hill, 2013.
3.	Ethem Alpaydin, Introduction to Machine Learning (Adaptive Computation and Machine Learning), The MIT Press, 2004.
Reference Books:	
1.	Elaine Rich, Kevin Knight and Shivashankar B. Nair, Artificial Intelligence, 3/e, McGraw Hill Education, 2008.
2.	Dan W. Patterson, Introduction to Artificial Intelligence and Expert Systems, PHI Learning, 2012.
e-Resources	
1.	https://www.tpointtech.com/artificial-intelligence-ai
2.	https://www.geeksforgeeks.org/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ME4103	PE	3	--	--	3	30	70	3 Hrs.
ADVANCED MATERIALS								
(For ME)								
Course Objectives: The student will acquire the knowledge								
1.	Introduce various metallic materials, their applications in extreme environments, and innovative forms like metallic foams.							
2.	Provide knowledge of natural and synthetic polymers, ceramics, their structures, processing, and industrial applications.							
3.	Explain the fundamentals, classifications, and reinforcements of composite materials and their engineering relevance.							
4.	Introduce shape memory alloys and functionally graded materials, focusing on their effects, properties, and applications.							
5.	Explore the principles, unique properties, and applications of nanomaterials compared to their bulk counterparts.							
Course Outcomes: At the end of the course student will be able to								
S.No	Outcome							Knowledge Level
1.	Apply material selection principles to choose suitable high-performance metals and alloys for aerospace and cryogenic applications.							K3
2.	Classify different types of ceramics based on their structure and functional properties.							K3
3.	Apply composite material concepts to choose suitable composite systems and reinforcement types for engineering uses.							K3
4.	Apply the principles of shape memory alloys (SMAs) and functionally graded materials (FGMs) in practical engineering applications.							K3
5.	Apply nanoscale principles to compare their behavior with bulk materials.							K3
SYLLABUS								
UNIT-I (10Hrs)	ADVANCED METALS & ALLOYS: Metallic materials: super alloys, titanium and nickel-based alloys and intermetallics, Materials for cryogenic applications, materials for space environment, Evaluation of materials for extreme environments, Introduction to metallic foams. Material selection using Asbhy charts.							
UNIT-II (10 Hrs)	CERAMICS: Applications- characteristics- classification, Processing of Ceramics-powder preparation- consolidation- hot compaction-drying, sintering,-finishing of Ceramics, Al ₂ O ₃ , Silicon nitrite Si ₃ N ₄ and Cermets, Areas of applications.							
UNIT-III (10 Hrs)	COMPOSITE MATERIALS: Introduction, polymer matrix composites, metal matrix composites, ceramic matrix composites, carbon-carbon composites, Fiber-reinforced composites and nature-made composite and applications. REINFORCEMENTS: Fibers-glass, silica, kevlar, carbon, boron, silicon carbide and boron carbide fibres.							

UNIT-IV (10 Hrs)	SHAPE MEMORY ALLOYS (SMA): Introduction-shape memory effect-Classification of shape memory alloys-composition-properties and applications of shape memory alloys. FUNCTIONALLY GRADED MATERIALS (FGM): Types of functionally graded materials-classification different systems- Preparation - Properties and applications of functionally graded materials.
UNIT-V (10 Hrs)	NANO MATERIALS: Introduction- Properties at nanoscales- Advantages and disadvantages applications in comparison with bulk materials (nano-structure, wires, tubes, and composites) state of art nano advanced-topic delivered by student.
Textbooks:	
1.	Nanomaterial – A.K. Bandyopadhyay / New age publishers.
2.	Material Science and Technology :A comprehensive treatment/Robert W. Cahn/VCH
3.	Engineering Mechanics of Composite Materials/Isaac and M Daniel/ Oxford University press.
Reference Books:	
1.	Mechanics of Composite Materials / R.M. Jones / Mc Graw Hill Company, New York, 1975.
2.	Analysis of Laminated Composite Structures / L.R. Calcote /Van Nostrand Rainfold, NY 1969.
3.	Analysis and Performance of Fibre Composites /B.D. Agarwal and L.J. Broutman/ Wiley-Inter science, New York, 1980.
4.	Mechanics of Composite Materials – Second Edition (Mechanical Engineering)/Autar K. Kaw/CRC Press.
e-Resources	
1.	https://youtu.be/VJwyo8WKU9I?si=PsVySoVIQ_0EAMBn

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ME4104	PE	3	--	--	3	30	70	3 Hrs.
SMART MANUFACTURING								
(For ME)								
Course Objectives:								
1.	Learn the basic ideas of smart and circular manufacturing and the challenges faced in modern industries.							
2.	Know how smart machines, sensors, and factory systems (CPS and 5C model) work together in manufacturing.							
3.	Get familiar with digital tools like Digital Twin, AI, ML, and IoT communication used for automation and connectivity.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Apply the principles of smart manufacturing, Industry 4.0, and Industry 5.0 by analyzing their evolution, stages, and the benefits they bring to modern manufacturing systems.							K3
2.	Apply the knowledge of smart sensors, subsystems, and interfaces used in IoT-based manufacturing, and describe their integration to enhance production systems.							K3
3.	Apply the 5C architecture of Cyber-Physical Systems (CPS) to explain the functioning of smart manufacturing systems, including PHM-based design, and distinguish between traditional factories and Industry 4.0 factories based on CPS implementation.							K3
4.	Apply digital twin implementation concepts in manufacturing and demonstrate the use of AI/ML for predictive maintenance to improve operational efficiency.							K3
5.	Apply an understanding of IoT network layers and protocols used in Industry 4.0 applications, and differentiate between their roles in enabling industrial automation.							K3
SYLLABUS								
UNIT-I (10Hrs)	CONCEPTS OF SMART MANUFACTURING: Definition and key characteristics of smart manufacturing, Corporate adaptation processes, manufacturing challenges, challenges vs technologies, Overview of circular manufacturing, Stages in smart manufacturing. Minimizing Six big losses in manufacturing with Industry 4.0, Industry 5.0, and their benefits. Evolution from industry 4.0 to industry 5.0 and industry 6.0, comparison of industry 4.0, industry 5.0 and industry 6.0, challenges and future scope.							
UNIT-II (10 Hrs)	SMART MACHINES AND SMART SENSORS: Concept and Functions of a Smart, Machine Salient features and Critical Subsystems of a Smart Machine, Smart sensors; smart sensors ecosystem, need, benefits and applications of sensors in industry, Sensing for Manufacturing Process in IIoT, Block Diagram of a IoT Sensing Device, Sensors in IIoT Applications, Smart Machine Interfaces							

UNIT-III (10 Hrs)	ARCHITECTURE OF CYBER-PHYSICAL SYSTEM (CPS): Functions of CPS, 5C Architecture; Smart Connection Level, Data-to- Information Level, Cyber Level, Cognition Level, Configuration Level. Design of PHM based CPS systems. Comparison of today's factory and Industry 4.0 factory by the implementation of 5C CPS architecture
UNIT-IV (10 Hrs)	DIGITAL TWIN: Introduction, applications of digital twins, impact zones of digital twins in manufacturing (factories/plants and OEMs), advantages of digital twins, basic steps of digital twin technology, Multiverse and AR and VR. MACHINE LEARNING (ML) AND ARTIFICIAL INTELLIGENCE (AI) IN MANUFACTURING: Introduction, benefits and applications of ML in industries, common approaches of ML; supervised and unsupervised, semi supervised and reinforced ML, Application of AI and ML for predictive maintenance.
UNIT-V (10 Hrs)	IOT CONNECTIVITY FOR INDUSTRY 4.0: Industrial communication requirement and its infrastructure, an overview of different types of networks, mesh network in industrial IoT, IoT protocols and the internet, TCP/IP (transmission control protocol/internet protocol) model, IoT connectivity standards: common protocols, application layer protocols, internet/network layer protocols, physical layer IoT protocols. Human-centric IOT connectivity in industry 5.0: Human – Machine collaboration, safe interaction between operators and connected systems.
Textbooks:	
1.	"Industry 4.0 The Industrial Internet of Things" by Alasdair Gilchrist, Apress
2.	"Industrial Internet of Things, Cyber Manufacturing System" by Sabina Jeschke, Christian Brecher, Houbing Song Danda B. Rawat, Springer
Reference Books:	
1.	"The Fourth Industrial Revolution" by Klaus Schwab
2.	"Internet of Things: A Hands-On Approach" by Arshdeep Bahga & Vijay Madisetti
3.	"Cyber-Physical Systems: From Theory to Practice" by Rajeev Alur, Daniel D. Lee, and Insup Lee
e-Resources	
1.	https://www.coursera.org/learn/industry-4-0
2.	https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-831-cyber-physical-systems-spring-2012/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ME4105	PE	3	--	--	3	30	70	3 Hrs.
MICRO ELECTRO MECHANICAL SYSTEMS								
(For ME)								
Course Objectives:								
1.	To illustrate the fundamentals of Micro Electro Mechanical Systems (MEMS) and demonstrate the use of micromachining and fabrication techniques in mechanical sensors and actuators.							
2.	To use the principles of thermal energy and heat transfer to analyze the operation of thermal sensors and actuators in MEMS devices.							
3.	To demonstrate the operating principles of Micro-Opto-Electro Mechanical Systems (MOEMS) and magnetic sensors and actuators for micro-scale sensing and actuation applications.							
4.	To analyze and illustrate the working of micro-fluidic systems used for fluid manipulation at micro scale.							
5.	To utilize the principles of chemical and biomedical microsystems for sensing, detection, and analytical applications.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Apply the fundamental concepts of MEMS fabrication processes such as micromachining, lithography, thin film deposition, doping and etching in the development of micro devices.							K3
2.	Apply the working principles of mechanical sensors and actuators used in MEMS applications.							K3
3.	Illustrate the operation of thermal sensors and actuators based on heat transfer principles.							K3
4.	Illustrate the principles of Micro-Opto-Electro Mechanical Systems (MOEMS) and magnetic sensors and actuators.							K3
5.	Apply the concepts of micro-fluidic systems and chemical/biomedical microsystems in sensing and analytical applications.							K3
SYLLABUS								
UNIT-I (10Hrs)	INTRODUCTION: Definition of MEMS, MEMS history and development, micromachining, lithography principles & methods, structural and sacrificial materials, thin film deposition, impurity doping, etching, surface micromachining, wafer bonding, LIGA. MECHANICAL SENSORS AND ACTUATORS: Principles of sensing and actuation: beam and cantilever, capacitive, piezo-electric, strain, pressure, flow, pressure measurement by micro phone, MEMS gyroscopes, shear mode piezo actuator, gripping piezo actuator, Inch worm technology.							
UNIT-II (10 Hrs)	THERMAL SENSORS AND ACTUATORS: Thermal energy basics and heat transfer processes, thermistors, thermo devices, thermo couple, micro machined thermo couple probe, Peltier effect heat pumps, thermal flow							

	sensors, micro hot plate gas sensors, MEMS thermo vessels, pyro electricity, shape memory alloys (SMA), U-shaped horizontal and vertical electro thermal actuator, thermally activated MEMS relay, micro spring thermal actuator, data storage cantilever.
UNIT-III (10 Hrs)	MICRO-OPTO-ELECTRO MECHANICAL SYSTEMS: Principle of MOEMS technology, properties of light, light modulators, beam splitter, micro lens, micro mirrors, digital micro mirror device (DMD), light detectors, grating light valve (GLV), optical switch, wave guide and tuning, shear stress measurement. MAGNETIC SENSORS AND ACTUATORS: Magnetic materials for MEMS and properties, magnetic sensing and detection, magneto resistive sensor, more on hall effect, magneto diodes, magneto transistor, MEMS magnetic sensor, pressure sensor utilizing MOKE(Magneto-Optical Kerr Effect), mag-MEMS actuators, bi-directional micro actuator, feedback circuit integrated magnetic actuator, large force reluctance actuator, magnetic probe based storage device.
UNIT-IV (10 Hrs)	MICRO FLUIDIC SYSTEMS: Applications, considerations on micro scale fluid, fluid actuation methods, dielectrophoresis (DEP), electro wetting, electro thermal flow, thermo capillary effect, electro osmosis flow, opto electro wetting (OEW), tuning using micro fluidics, typical micro fluidic channel, micro fluid dispenser, micro needle, molecular gate, micro pumps. RADIO FREQUENCY (RF) MEMS: RF-based communication systems, RF MEMS, MEMS inductors, tuner/filter, resonator, clarification of tuner, filter, resonator, MEMS switches, phase shifter.
UNIT-V (10 Hrs)	CHEMICAL AND BIOMEDICAL MICRO SYSTEMS: Sensing mechanism & principle, membrane-transducer materials, chem-lab-on-a-chip (CLOC), chemo-resistors, chemo-capacitors, chemo-transistors, electronic nose (E-nose), mass sensitive chemo-sensors, fluorescence detection, calorimetric spectroscopy.
Textbooks:	
1.	MEMS, Nitaigour Premchand Mahalik, TMH Publishing co.
Reference Books:	
1.	Foundation of MEMS, Chang Liu, Prentice Hall Ltd.
2.	MEMS and NEMS, Sergey Edward Lyshevski, CRC Press, Indian Edition.
3.	MEMS and Micro Systems: Design and Manufacture, Tai-Ran Hsu, TMH Publishers.
4.	Introductory MEMS, Thomas M. Adams, Richard A. Layton, Springer International Publishers.
e-Resources	
1.	https://nptel.ac.in/courses/117105082
2.	https://nptel.ac.in/courses/108108113

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ME4106	PE	3	--	--	3	30	70	3 Hrs.
FUNCTIONAL MATERIALS AND APPLICATIONS								
(For ME)								
Course Objectives:								
1.	To introduce the concepts of crystal structures, defects, and deformation mechanisms in materials.							
2.	To provide understanding of properties and applications of various materials.							
3.	To learn the concepts of semiconductor physics and charge transport mechanisms.							
4.	To obtain knowledge on quantum phenomena in nano electronic materials and their applications.							
Course Outcomes: At the end of the Course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Illustrate the concepts of crystal structures, crystallographic directions and planes, crystal defects, slip systems, and phase changes in materials							K3
2.	Explain the concepts of electrical and magnetic properties of materials using classical and quantum theories.							K3
3.	Explain the concepts of intrinsic and extrinsic semiconductors and charge transport mechanisms.							K3
4.	Describe the optical behavior of materials and the operation of optoelectronic devices							K3
5.	Discuss the important phenomena associated with nano electronic materials.							K3
SYLLABUS								
UNIT-I (8 Hrs)	CRYSTALLOGRAPHY: Crystal structures: BCC, FCC and HCP – directions and planes - linear and planar densities – crystal imperfections- edge and screw dislocations – grain and twin boundaries - Burgers vector and elastic strain energy- Slip systems, plastic deformation of materials - Polymorphism – phase changes.							
UNIT-II (8 Hrs)	ELECTRICAL AND MAGNETIC PROPERTIES OF MATERIALS: Classical free electron theory - Expression for electrical conductivity – Thermal conductivity, expression - Quantum free electron theory: Tunneling – degenerate states – Fermi- Dirac statistics – Density of energy states – Electron in periodic potential – Energy bands in solids – tight binding approximation - Electron effective mass – concept of hole. Magnetic materials: Dia, para and ferromagnetic effects – paramagnetism in the conduction electrons in metals – exchange interaction and ferromagnetism – quantum interference devices –GMR devices							
UNIT-III (12 Hrs)	SEMICONDUCTORS AND TRANSPORT PHYSICS: Intrinsic Semiconductors – Energy band diagram – direct and indirect band gap semiconductors – Carrier concentration in intrinsic semiconductors – extrinsic semiconductors - Carrier concentration in N-type & P-type semiconductors – Variation of carrier concentration with temperature – Carrier transport in Semiconductors: Drift, mobility and diffusion – Hall							

	effect and devices – Ohmic contacts – Schottky diode
UNIT-IV (10Hrs)	OPTICAL PROPERTIES OF MATERIALS: Classification of optical materials – Optical processes in semiconductors: optical absorption and emission, charge injection and recombination, optical absorption, loss and gain. Optical processes in quantum wells – Optoelectronic devices: light detectors and solar cells – light emitting diode – laser diode - optical processes in organic semiconductor devices –excitonic state – Electro-optics and nonlinear optics: Modulators and switching devices – plasmonics.
UNIT-V (10Hrs)	NANOELECTRONIC MATERIALS: Quantum confinement – Quantum structures – quantum wells, wires and dots – Zener-Bloch oscillations – Resonant tunneling – quantum interference effects - mesoscopic structures – Single electron phenomena – Single electron Transistor. Semiconductor photonic structures – 1D, 2D and 3D photonic crystal. Active and passive optoelectronic devices – photo processes – spintronics – carbon nanotubes: Properties and applications
Textbooks:	
1.	V.Raghavan. Materials Science and Engineering: A First Course, Prentice Hall India Learning Private Limited, 2015.
2.	S.O. Kasap, Principles of Electronic Materials and Devices, Mc-Graw Hill, 2018
3.	Jasprit Singh, Semiconductor Devices: Basic Principles, Wiley (India), 2007.
4.	Jasprit Singh, Semiconductor Optoelectronics: Physics and Technology, Mc-Graw Hill India (2019)
5.	G.W.Hanson. Fundamentals of Nanoelectronics. Pearson Education (Indian Edition), 2009.
Reference Books:	
1.	Callister's Materials Science and Engineering. Wiley (Indian Edition), 2014.
2.	Wendelin Wright and Donald Askeland, Essentials of Materials Science and Engineering, CL Engineering, 2013.
3.	Robert F.Pierret, Semiconductor Device Fundamentals, Pearson, 2006
4.	Pallab Bhattacharya, Semiconductor Optoelectronic Devices, Pearson, 2017
5.	Ben Rogers, Jesse Adams and Sumita Pennathur, Nanotechnology: Understanding Small Systems, CRC Press, 2017
e-Resources	
1.	https://nptel.ac.in/courses/115105134

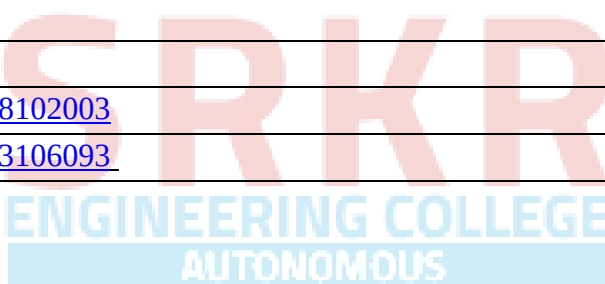
Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ME4108	PE	3	--	--	3	30	70	3 Hrs.
HYDROGEN AND FUEL CELL TECHNOLOGY								
(For ME)								
Course Objectives:								
1.	To introduce the fundamental concepts of hydrogen as an energy carrier and its role in a sustainable energy system.							
2.	To provide comprehensive knowledge on various hydrogen production technologies.							
3.	To impart understanding of hydrogen storage technologies and associated challenges.							
4.	To explore various applications of hydrogen in energy conversion devices and engines.							
5.	To explain the principles, types, and performance characteristics of fuel cells.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Apply the physical and chemical properties of hydrogen to determine specific infrastructure requirements for safe production and transport.							K3
2.	Illustrate the steps of hydrogen production and calculate the energy requirements for thermal, electrochemical, and biological hydrogen production.							K3
3.	Compute the storage capacity, pressure requirements, and energy density for compressed, cryogenic, and solid-state systems.							K3
4.	Illustrate the integration of hydrogen into IC engines and hybrid systems.							K3
5.	Compute the performance of different fuel cells and solve basic problems comparing them to batteries.							K3
SYLLABUS								
UNIT-I (10Hrs)	INTRODUCTION OF HYDROGEN ENERGY SYSTEMS: Properties of hydrogen as fuel, Hydrogen pathways introduction-current uses, general introduction to infrastructure requirement for hydrogen production, storage, dispensing, utilization, safety and hydrogen production plants.							
UNIT-II (10 Hrs)	HYDROGEN PRODUCTION PROCESSES: Thermal-Steam reformation, thermo chemical water splitting, gasification-pyrolysis, nuclear thermal catalytic and partial oxidation methods, Electrochemical-Electrolysis, photo electro chemical, Biological-Anaerobic digestion, fermentation microorganism, PM based electrolyser.							
UNIT-III (10 Hrs)	HYDROGEN STORAGE: Physical and chemical properties, general storage methods, compressed storage-composite cylinders, glass micro sphere storage, zeolites, metal hydride storage, chemical hydride storage and cryogenic storage, carbon-based materials for hydrogen storage.							
UNIT-IV (10 Hrs)	HYDROGEN UTILIZATION: Overview of hydrogen utilization, IC Engines, gas turbines, hydrogen burners, power plant, domestic cooking gas, marine applications,							

	hydrogen dual fuel engines.
UNIT-V (10 Hrs)	FUEL CELLS: History – principle - working - thermodynamics and kinetics of fuel cell process –performance evaluation of fuel cell – comparison on battery Vs fuel cell, Types of fuel cells – AFC, PAFC, SOFC, MCFC, DMFC, PEMFC, microbial fuel cells, relative merits and demerits and application of fuel cells.
Textbooks:	
1.	Sorenson B, Hydrogen and Fuel Cells: Emerging Technologies and Applications, Bent Sorenson, Academic Press (2005).
2.	Hordeski MF, Hydrogen and Fuel Cells: Advances in Transportation and Power, The Fairmont Press, Inc. (2009)
Reference Books:	
1.	Hydrogen Fuel: Production, Transport, and Storage- Ram B. Gupta
2.	Hydrogen Production, Storage, and Utilization (2025)A. Tcharkhtchi, et al.
3.	Busby RL, Hydrogen and Fuel Cells: A Comprehensive Guide, Penn Well Books (2005).
e-Resources	
1.	https://onlinecourses.nptel.ac.in/ NOC25_CH93
2.	https://nptel.ac.in/courses/103102015



Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23ME4109	PE	3	--	--	3	30	70	3 Hrs.
NANO TECHNOLOGY								
(For ME)								
Course Objectives:								
1.	This course introduces the fundamentals of nano-scale engineering and manufacturing							
2.	This course gives a detailed understanding of various applications of nano technology.							
3.	Well-established and novel synthesis/fabrication methods nanostructures will be discussed giving a broad overview of nanomanufacturing processes.							
4.	Standard characterization methods will be elucidated using various examples							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Apply the principles of nanoscience to classify nanomaterials and interpret their basic crystallographic parameters using Bragg's law.							K3
2.	Interpret the influence of nanoscale on material properties.							K3
3.	Demonstrate a comprehensive understanding of nano fabrication methods.							K3
4.	Utilize various characterization methods to understand the nanomaterials.							K3
5.	Illustrate the structure, synthesis, and applications of carbon-based nanomaterials.							K3
SYLLABUS								
UNIT-I (8Hrs)	Introduction: History of nano science, definition of nano meter, nano materials, nano technology. Classification of nano materials. Bragg's equation, Crystal symmetries, crystal directions, crystal planes. Band structure.							
UNIT-II (10Hrs)	Properties of nano materials: Mechanical properties, electrical properties, dielectric properties, thermal properties, magnetic properties, opto electronic properties. Effect of size reduction on properties, electronic structure of nano materials.							
UNIT-III (12Hrs)	Synthesis and Fabrication: Synthesis of bulk polycrystalline samples, growth of single crystals. Synthesis techniques for preparation of nano particle Bottom-Up Approach – sol gel synthesis, hydrothermal growth, thin film growth, PVD and CVD; Top Down Approach – Ball milling, micro fabrication, lithography. Requirements for realizing semiconductor nano structures, growth techniques for nano structures.							
UNIT-IV (12Hrs)	Imaging/characterization: Practice state-of-the-art characterization methods for nanomaterials, understanding and critiquing nanomaterial safety and handling methods required during characterization, Imaging/characterization of nanostructures General considerations for imaging, Scanning probe techniques: XRD, SEM, TEM, AFM and NSOM, photoluminescence spectra, Raman spectroscopy.							

UNIT-V (8Hrs)	Carbon Nano Technology: Characterization of carbon allotropes, synthesis of diamond – nucleation of diamond, growth and morphology, synthesis of Quantum dots. Applications of nano crystalline diamond films, graphene, applications of carbon nano tubes. Applications Of Nano Technology: Applications in material science, bioengineering, biology and medicine, surface science, sensors, energy, and environment. Applications of nano structured thin films, applications of quantum dots.
Textbooks:	
1.	Nano science and nano technology by M.S Ramachandra Rao, Shubra Singh, Wiley publishers.
2.	Scanning Electron Microscopy and X-Ray Microanalysis by Joseph Goldstein Lyman, Dale E. Newbury, David C. Joy, Patrick Echlin, Springer.
Reference Books:	
1.	Introduction to Nano Technology by Charles P. Poole, Jr., Frank J.Owens, Wiley publishers.
2	Nanotechnology by Jermy J Ramsden, Elsevier publishers
3	Nano Materials- A.K.Bandyopadhyay/ New Age Introdu.
4	Nano Essentials- T.Pradeep/TMH.
5	Nanotechnology the Science of Small by M.A Shah, K.A Shah, Wiley Publishers.
6	Principles of Nanotechnology by Phani Kumar, Scitech.
e-Resources	
1.	https://nptel.ac.in/courses/118102003
2.	https://nptel.ac.in/courses/113106093



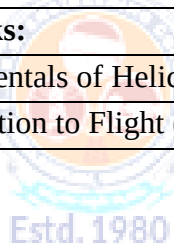
Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ME4110	PE	3	--	--	3	30	70	3 Hrs.
CONVENTIONAL AND FUTURISTIC VEHICLE TECHNOLOGY								
(For ME)								
Course Objectives:								
1.	To study the advanced engine technologies							
2.	To learn various advanced combustion technologies and its benefits.							
3.	To learn the methods of using low carbon fuels and its significance.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Apply the working principles of advanced engine technologies.							K3
2.	Illustrate the operational differences between conventional combustion and advanced low-temperature combustion concepts in reducing emissions							K3
3.	Demonstrate the characteristics, adoption challenges, and applications of various low-carbon alternatives like ammonia, methane, and hydrogen fuels							K3
4.	Demonstrate the energy flow and operational modes within conventional hybrids, modern hybrids, and pure electric vehicle configurations.							K3
5.	Summarize the functionality of automotive fuel cell control systems and the associated onboard liquid and compressed hydrogen storage methods							K3
SYLLABUS								
UNIT-I (10Hrs)	ADVANCED ENGINE TECHNOLOGY: Gasoline Direct Injection, Common Rail Direct Injection, Variable Compression Ratio Turbocharged Engines, Electric Turbochargers, VVT, Intelligent Cylinder De-activation, After Treatment Technologies, Electric EGR, Current EMS architecture.							
UNIT-II (10 Hrs)	COMBUSTION TECHNOLOGY : SI engine combustion, CI engine Combustion, Conventional Dual Fuel Combustion, Low Temperature Combustion Concepts Controlled Auto Ignition, Homogeneous Charge Compression Ignition, Premixed Charge Compression Ignition, Partially Premixed Compression Ignition, Reactivity Controlled Compression Ignition, Gasoline Direct Injection Compression Ignition.							
UNIT-III (10 Hrs)	LOW CARBON FUEL TECHNOLOGY : Alcohol Fuels, Ammonia Fuel and Combustion, Methane Technology, Dimethyl Ether, Hydrogen Fuel Technology, Challenges and way forward							
UNIT-IV (10 Hrs)	HYBRID AND ELECTRIC VEHICLE (BATTERY POWERED) : Conventional Hybrids (Conventional ICE + Battery), Modern Hybrids (RCCI/GDCI Engine + Battery), Pure Electric Vehicle Technology - Challenges and Way forward							
UNIT-V	FUEL CELL TECHNOLOGY: Fuel cells for automotive applications Technology							

(10 Hrs)	advances in fuel cell vehicle systems Onboard hydrogen storage Liquid hydrogen and compressed hydrogen Metal hydrides, Fuel cell control system - Alkaline fuel cell - Road map to market.
Textbooks:	
1.	Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press, 200
2.	Rakesh Kumar Maurya, Characteristics and Control of Low Temperature Combustion Engines. ISBN 978-3-319-68507-6, SPRINGER
Reference Books:	
1.	Iqbal Hussein, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, 2003.
2.	James Larminie, John Lowry, Electric Vehicle Technology Explained, Wiley, 2003
3.	Rand D.A.J, Woods, R & Dell RM Batteries for Electric vehicles, John Wiley & Sons, 1998
e-Resources	
1.	https://nptel.ac.in/courses/112103617
2.	https://nptel.ac.in/courses/103102015



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ME4111	PE	3	--	--	3	30	70	3 Hrs.
AUTONOMOUS SYSTEMS								
(For ME)								
Course Objectives:								
1.	To Introduce the concepts of intelligent agents and multi-agent systems in the context of autonomous systems.							
2.	To Provide an understanding of the architectural design and autonomy levels in unmanned systems like AGVs, AUVs, and drones.							
3.	To Equip with knowledge of modelling and control techniques for different types of unmanned ground, aerial, and underwater vehicles.							
4	To Familiarize with various sensors, actuators, and computing platforms used in autonomous systems.							
5	To develop a foundational understanding of aerial robot flight concepts, control strategies, motion planning, and communication protocols.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Apply the structure and functioning of intelligent agents and multi-agent systems for collaborative autonomous tasks.							K3
2.	Apply and differentiate the architectural features and levels of autonomy in ground, aerial, and underwater autonomous systems.							K3
3.	Apply control techniques to different types of unmanned vehicles considering dynamics and motion constraints.							K3
4.	Identify suitable sensors and actuators used in autonomous systems, and implement basic sensor fusion techniques.							K3
5.	Apply control algorithms and motion planning strategies for aerial robots including trajectory tracking, guidance, and navigation.							K3
SYLLABUS								
UNIT-I (10Hrs)	INTRODUCTION TO AGENT SYSTEMS Introduction to Agents, Agent Architectures, and Multi-agent Systems and Society of Agents, Distributed Problem Solving and Planning, Case Study: Collaborative Robotics, Robocup.							
UNIT-II (10 Hrs)	ARCHITECTURE OF AUTONOMOUS SYSTEMS Degree of Autonomy, Reactive Systems, Real-time Systems, Architecture of AGV's, AUV, Drones, Tele-Operation, AR, VR applications.							
UNIT-III (10 Hrs)	CLASSIFICATION, MODELLING, AND CONTROL ASPECTS OF UNMANNED VEHICLES Different types of unmanned vehicles: ground (wheeled and legged), aerial (fixed, flapping, and rotary wings), underwater Drones, Modelling of unmanned vehicles							

	considering basic forces, kinematics, and dynamics, Discussion on different types of control for aerial, underwater (fins and propulsion control), ground (biped and quadruped motion control for legged robots).
UNIT-IV (10 Hrs)	SENSORS AND ACTUATORS Types of sensors used in unmanned vehicles (proximity, IMU, magnetometers, thermal imaging, vision, LiDAR, GPS, RTK, etc.) and their characteristics, Sensor data aggregation, processing, and sensor fusion, Introduction to popular computing platforms for data processing, Types of actuators: motors, servos, harmonic drive, linear actuators.
UNIT-V (10 Hrs)	FLIGHT CONCEPTS Basic Aerial Robot Flight Concepts, Micro-aerial vehicle, Frame Rotations and Representations, Aerial robots equations of motion, State-Space Form, Time, Motion, and Trajectories, Linearization, 2-D and 3-D control of Aerial robots, PID Control, LQR control, Motion planning, Collision-free Navigation, Sensing and Estimation, Vision-based Guidance for aerial robots, Introduction to Communication and Networking Protocols.
Textbooks:	
1.	Introduction to Autonomous Mobile Robots by Roland Siegwart, Illah Reza Nourbakhsh, and Davide Scaramuzza (2018), MIT Press
2.	Multiagent System (Second Edition) edited by Gerhard Weiss (2013), MIT Press
Reference Books:	
1.	Fundamentals of Helicopter Dynamics (1st Edition) by C. Venkatesan (2014), CRC Press
2.	Introduction to Flight (8th Edition) by John D. Anderson (2015), McGraw-Hill Education



SREE NARAYANA
ENGINEERING COLLEGE
AUTONOMOUS

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ME4113	PC	--	--	2	1	30	70	3 Hrs.
CAD/CAM LAB								
(For ME)								
Course Objectives:								
1	Introduce students to simulation-based analysis of structural, thermal, and vibrational behavior using FEA tools.							
2	Provide practical knowledge of CNC part programming for lathe and milling Operations.							
3	Familiarize students with CAM software for automatic toolpath planning and G-code Generation.							
4	Introduce the concepts and workflow of rapid prototyping using 3D printing machines.							
5	Enable students to simulate and visualize the 3D printing process through online virtual labs.							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1	Analyze structural and thermal engineering problems using CAE software to determine stresses, deflections, natural frequencies, harmonic response, heat transfer characteristics, and buckling behavior of engineering components.							K4
2	Evaluate and create CNC part programs for lathe and milling operations, including turning, facing, circular interpolation, and pocket milling, while demonstrating proficiency in automated tool path generation and G-code creation using CAM software.							K4
3	Analyze the rapid prototyping process and interpret the workflow involved in creating simple parts using RP machines and virtual 3D printing simulations through the Virtual Labs platform.							K4
SYLLABUS								
1	Experiments to determine stresses, deflection, natural frequencies, harmonic analysis, HT analysis and buckling analysis. a) Determination of deflection and stresses in 2D and 3D trusses and beams. b) Determination of principal and Von-mises stresses in plane stress, plane strain and axisymmetric components. c) Determination of stresses in 3D and shell structures (at least one example in each case) d) Estimation of natural frequencies and mode shapes, harmonic response of 2D beam. e) Steady state heat transfer analysis of plane and axisymmetric components. f) Buckling analysis							

2	Study of CNC part programming fundamentals and write part programmes for simple components on CNC lathe and Mill and Study of RP machine. A. CNC part programming for turned components (i) Plain turning and facing (ii) Step Turning Operation (iii) Taper turning B. CNC programming for milled components (i) Circular interpolation (ii) End milling (iii) Pocket milling
3	Automated CNC Tool path and G-Code generation using CAM packages.
4	Study and demonstration of RP machine-creation of simple parts.
5	Virtual 3D Printing Simulation lab using Vlabs.
Reference Books:	
1	CAD/CAM Theory and Practice by Ibrahim Zeid.
2	CAD/CAM Principles and Applications by P.N. Rao, Tata McGraw Hill Publishing Company Ltd.
3	Computer Integrated Design and Manufacturing by David D. Bedworth, Mark R. Henderson, Philip M. Wolfe.



Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ME4114	SEC	--	1	2	2	30	70	3 Hrs.
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING LAB								
(For ME)								
Course Objectives:								
1	Students will acquire the knowledge of artificial intelligence and machine learning models using various software tools.							
2	To enable the students write coding for various artificial intelligence and machine learning algorithms.							
Course Outcomes: Students at the end of the course will be able to								
S.No	Outcome							Knowledge Level
1	Apply various Python libraries.							K3
2	Apply programming techniques for regression methods.							K3
3	Apply clustering and basic machine learning techniques for data analysis.							K3
4	Analyze engineering datasets for prediction and classification problems.							K4
5	Analyze and implement computer vision and deep learning techniques (OpenCV and CNN) for image processing and defect detection applications.							K4
SYLLABUS								
1	Learning of Python libraries – NumPy, Pandas, Matplotlib, Seaborn.							
2	Numerical examples on Python libraries and data visualization.							
3	Data preprocessing and data cleaning using Python.							
4	Write a program for Linear Regression.							
5	Write a program for Logistic Regression.							
6	Write a program for K-Nearest Neighbour (KNN).							
7	Write a program to build a Decision Tree.							
8	Write a program to build a Naïve Bayes classifier.							
9	Write a program for Support Vector Machine (basic implementation).							
10	Write a program for K-Means clustering.							
11	Write a program for basic image processing and defect detection using OpenCV.							
12	Write a program for Convolutional Neural Network (CNN) for image classification (basic implementation).							
Reference Books:								
1	“Python for Data Analysis” By Wes McKinney							
2	“Machine Learning” By Tom M. Mitchell							
NOTE: Database can be taken from https://www.kaggle.com/DATABASE								

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC4101	MC	2	--	--	--	30	--	--
CONSTITUTION OF INDIA								
(Common to all)								
Course Objectives:								
1.	To provide understanding of the fundamental principles, sources, and key features of the Indian Constitution.							
2.	To familiarize students with the structure and functioning of Union, State, and Local Governments in India.							
3.	To develop awareness about the roles of constitutional bodies like the Election Commission and welfare commissions in promoting democracy and social justice.							
Course Outcomes: At the end of the course, students will be able to								
S.No.	Outcome							Knowledge Level
1.	Apply the concepts of the Indian Constitution to interpret its features such as Citizenship, Preamble, Fundamental Rights and Duties, and Directive Principles of State Policy.							K3
2.	Apply the structure and functions of the Union Government to analyze the roles of executive, legislature, and judiciary in the Indian administrative system.							K3
3.	Apply the structure and functions of the State Government to interpret the roles of the Governor, Chief Minister, and State Secretariat in administration.							K3
4.	Apply the structure and functions of local administration to interpret the roles of district, municipal, and Panchayati Raj institutions in grassroots governance.							K3
5.	Apply the roles and functions of the Election Commission and various welfare commissions to interpret their significance in ensuring democratic governance and social justice.							K3
SYLLABUS								
UNIT-I (6 Hrs)	Introduction to Indian Constitution: Constitution meaning of the term, Indian Constitution - Sources and constitutional history, Features - Citizenship, Preamble, Fundamental Rights and Duties, Directive Principles of State Policy.							
UNIT-II (6 Hrs)	Union Government and its Administration Structure of the Indian Union: Federalism, Centre- State relationship, President: Role, power and position, PM and Council of ministers, Cabinet and Central Secretariat, Lok Sabha, Rajya Sabha, The Supreme Court and High Court: Powers and Functions.							
UNIT-III (6 Hrs)	State Government and its Administration Governor - Role and Position - CM and Council of ministers, State Secretariat: Organisation, Structure and Functions.							

UNIT-IV (6 Hrs)	Local Administration - District's Administration Head - Role and Importance, Municipalities — Mayor and role of Elected Representative - CEO of Municipal Corporation Pachayati Raj: Functions PRI: Zilla Panchayat, Elected officials and their roles, CEO Zilla Panchayat: Block level Organizational Hierarchy - (Different departments), Village level - Role of Elected and Appointed officials - Importance of grass root democracy.
UNIT-V (6 Hrs)	Election Commission: Election Commission- Role of Chief Election Commissioner and Election Commissionerate State Election Commission: Functions of Commissions for the welfare of SC/ST/OBC and women.
Textbooks:	
1.	Durga Das Basu, Introduction to the Constitution of India, Prentice — Hall of India Pvt. Ltd..New Delhi.
2.	Subash Kashyap, Indian Constitution, National Book Trust.
Reference Books:	
1.	J.A. Siwach, Dynamics of Indian Government & Politics, Sterling Publishers.
2.	D.C. Gupta, Indian Government and Politics, S Chand.
3.	H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes, Universal Law Publication.
4.	M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice Hall of India Pvt. Ltd.. New Delhi.
5.	Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Challenges to Civil Rights Guarantees in India, Oxford University Press 2012.
Estd. 1980 AUTONOMOUS	
e-Resources	
1.	https://onlinecourses.swayam2.ac.in/e-learning/preview/imb26_mg151
2.	https://onlinecourses.nptel.ac.in/noc24_lw05/preview
3.	https://nptel.ac.in/courses/129106750



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

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SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		IV / IV - B.Tech. II - Semester							
MECHANICAL ENGINEERING									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23ME4201	Project and Internship	PR	--	--	24	12	60	140	200
TOTAL			--	--	24	12	60	140	200



SRKR

ENGINEERING COLLEGE

AUTONOMOUS

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23ME4201	PR	--	--	24	12	60	140	3 Hrs.
PROJECT WORK								
(For ME)								
Course Objectives:								
1	To provide an opportunity to work in group on a topic / problem / experimentation							
2	To encourage creative thinking process							
3	To provide an opportunity to analyze and discuss the results to draw conclusions							
4	To acquire and apply fundamental principles of planning and carrying out the work plan of the project through observations, discussions and decision-making process.							
Course Outcomes: At the end of the course the students will be able to								
S.No.	Outcome							Knowledge Level
1	Identify and define engineering problems using systematic literature survey techniques.							K3
2	Formulate research objectives and develop suitable research methodologies for solving the identified engineering problems							K4
3	Conduct experiments, analyse and interpret results, and propose suitable solutions for engineering problems.							K5
4	Work in multi-disciplinary teams and provide solutions to problems that arise in multi- disciplinary work with professional and ethical responsibility.							K4
5	Communicate effectively through written reports, oral presentations, and discussion.							K4
*The object of Project Work is to enable the student to take up investigative study in the broad field of Mechanical Engineering, either fully theoretical/practical or involving both theoretical and practical work to be assigned by the Department on an individual basis or a group of students, under the guidance of a Supervisor. This is expected to provide a good initiation for the student(s) in R&D work.								
The assignment to normally include:								
a) Survey and study of published literature on the assigned topic.								
b) Working out a preliminary approach to the problem relating to the assigned topic.								
c) Conducting preliminary Analysis/Modeling/Simulation/Experiment/Design/ Feasibility.								
d) Preparing a written report on the study conducted for presentation to the department.								
e) Final Seminar, as oral Presentation before a departmental committee.								